

Ernesto Carlton

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Aerospace Engineering Student

Projects

Goose Deterrent System

Aug 2024 – Present

- Led cross-functional team to design and prototype acoustic deterrent system.
- Managed project timeline, parts sourcing, and weekly design reviews.

AFC Piezo-Actuator

March 2025 – Present

- Characterized piezo actuator response under varying load conditions.
- Built MATLAB scripts to process sensor data and plot frequency response.
- Collaborated on PCB layout and firmware integration for control board.

Wind Turbine Revitalization Project

Jan 2025 – May 2025

- Assessed turbine electrical subsystem and identified refurbishment scope.
- Supported wiring diagrams and safety checklist for lab re-commissioning.

Skills

CAD/Simulation:	SolidWorks	ANSYS	CATIA	Fusion 360	MATLAB Simulink	LabVIEW
Programming:	Python	C++	MATLAB	Java	SQL	VBA
Embedded/Systems:	Arduino	Raspberry Pi	STM32	I2C/SPI	RTOS basics	PCB design

Work Experience

Mostardi Platt

May 2023 – Aug 2024

Air Emissions Engineer

Elmhurst, IL

- Prepared air emissions reports and compliance documentation.
- Performed stack testing data review and QA for client deliverables.
- Supported field engineers with calculations and regulatory research.

GreenWay Landscaping

May 2025 – Dec 2025

Operations Lead

Aurora, IL

- Supervised daily crew operations and equipment maintenance schedules.

Education

Illinois Institute of Technology

May 2027

Bachelor of Science in Aerospace Engineering

Chicago, IL

Benedictine University

May 2027

Bachelor of Arts in Engineering Science

Lisle, IL